IndiGo

Airlines Booking Data Analysis

Project Overview

IndiGo is one of India's leading airlines with a strong domestic and international network. This project analyses Indigo Airlines' flight booking data to uncover operational, revenue, and customer behavior insights that support data-driven decision-making and business growth.

Business Problem

With increasing competition in the aviation industry, Indigo Airlines needs to optimise flight operations, improve customer satisfaction, and maximise route profitability. The objective of this analysis is to identify key trends and operational gaps that can help improve efficiency, occupancy, and overall revenue performance.

Data Set Overview

The dataset contains Indigo Airlines' flight booking and operational data, covering details related to routes, passengers, bookings, revenue, occupancy, and flight performance. The data includes both categorical and numerical variables related to bookings, routes, travel preferences, and flight operations.

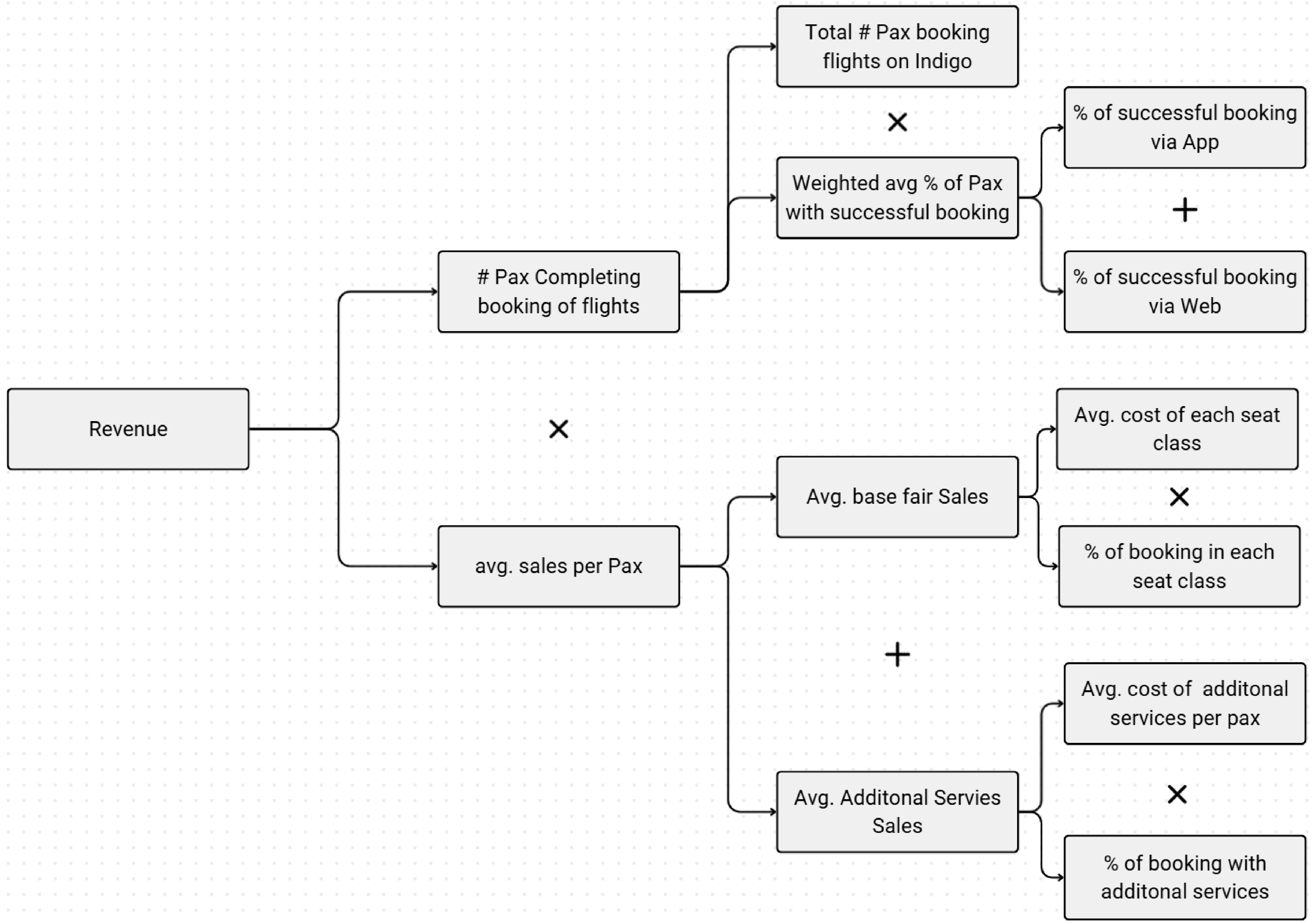
Fields	Data Type	Category	Description
Num Passengers	Numeric	Discrete	Number of passengers booked
Sales Channel	Categorical	Nominal	Booking platform/channel
Trip Type	Categorical	Nominal	One-way or round-trip
Purchase Lead	Numeric	Discrete	Days between booking and flight
Length of Stay	Numeric	Discrete	Total stay duration in days
Flight Hour	Qualitative	Time-based	Departure hour
Flight Day	Categorical	Ordinal	Day of travel
Route	Categorical	Nominal	Flight route information
Booking Origin	Categorical	Nominal	Passenger booking location
Wants Extra Baggage	Categorical	Binomial	Baggage preference
Wants Preferred Seat	Categorical	Binomial	Seat preference
Wants In-flight Meals	Categorical	Binomial	Meal preference
Flight Duration Planned	Numeric	Continuous	Scheduled flight duration
Flight Duration Actual	Numeric	Continuous	Actual flight duration
Booking Complete	Categorical	Binomial	Booking completion status
Seat Class	Categorical	Ordinal	Economy, Premium Economy, Business
Booking Amount	Numeric	Discrete	Total booking revenue

User Journey Map

The user journey map illustrates the complete flight booking process from flight search to booking confirmation. This framework was used to understand customer interaction points, identify operational touchpoints, and map business metrics across different stages of the booking funnel.

Search (App/Web) → Trip Selection → Flight Preference → Flight Selection → Pax Details → Add -on Page (Additonal Services) → Payment Page → Booking Confirmed

Metric Tree



Revenue
Pax Completing booking of flights
×
avg. sales per Pax
Total # Pax booking flights on Indigo
×
Weighted avg % of Pax with successful booking
% of successful booking via App
+
% of successful booking via Web
Avg. base fair Sales
+
Avg. Additonal Servies Sales
Avg. cost of each seat class
×
% of booking in each seat class
Avg. cost of additonal services per pax
×
% of booking with additonal services

KPI Dashboard

KPIs were designed to evaluate booking performance, customer preferences, revenue contribution, and operational efficiency.

15.43%

BOOKING COMPLETION RATE

45.82%

BOOKINGS WITH ADDITIONAL SERVICES

3.66K

AVERAGE BOOKING AMOUNT

91.78%

BOKINGS THROUGH INTERNET CHANNEL

~33.4%

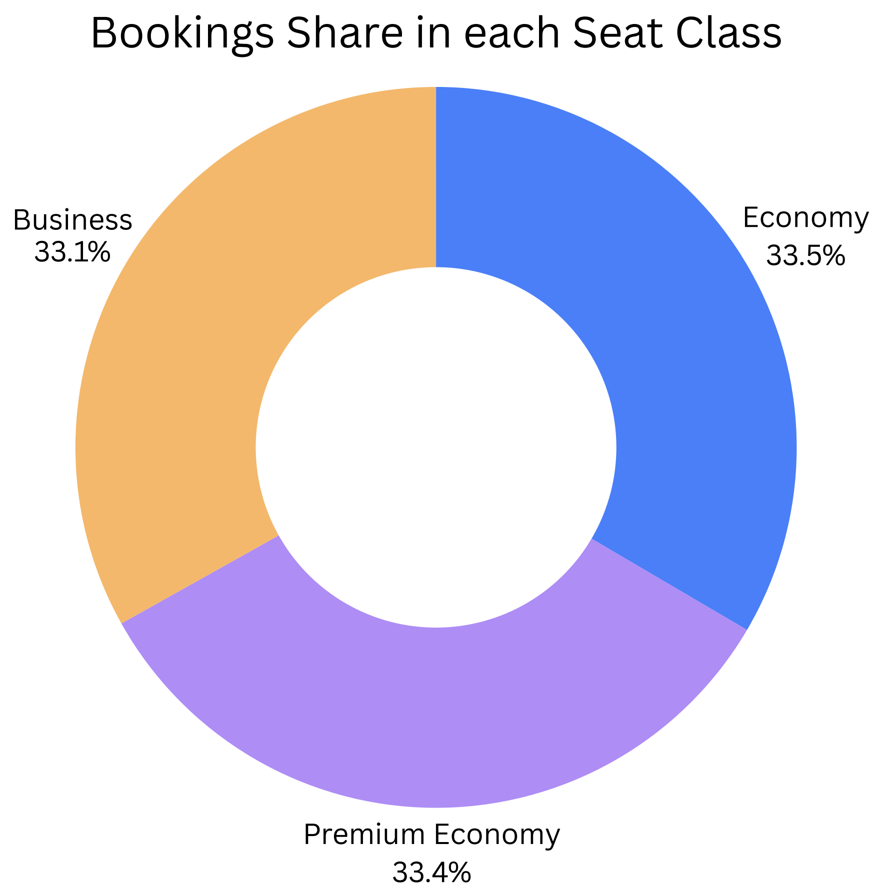
BOKINGS IN EACH SEAT CLASS

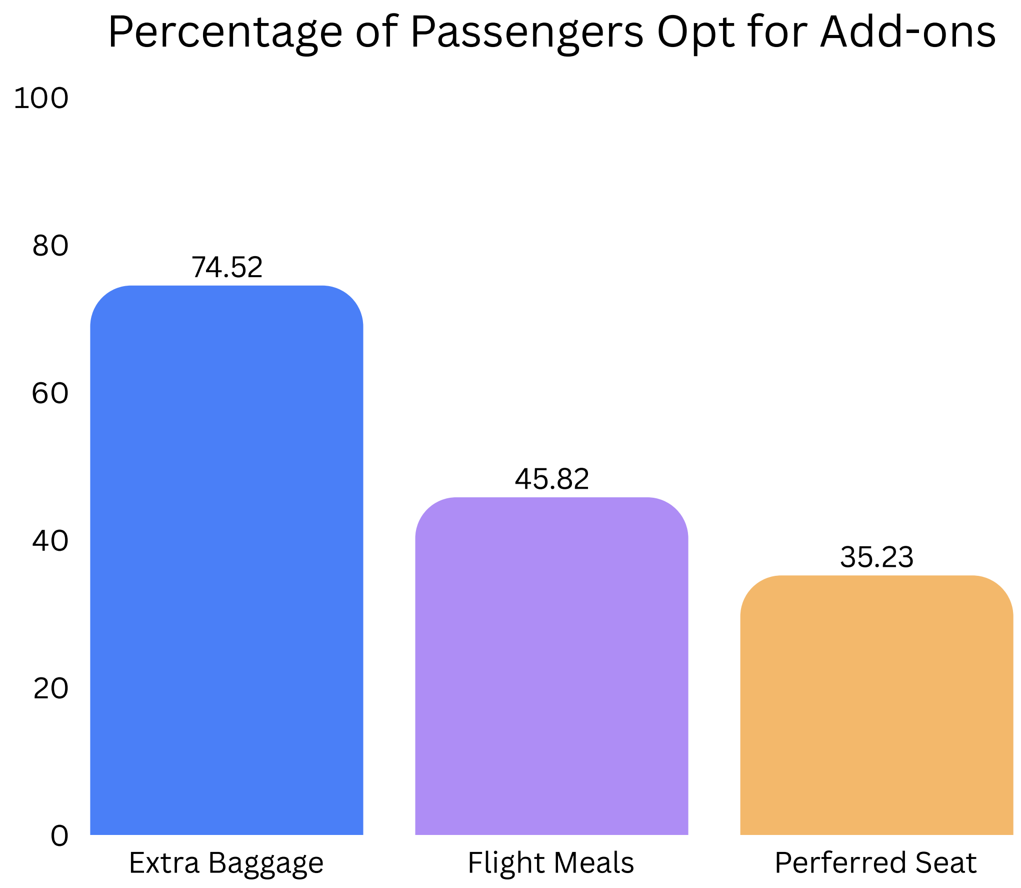
Hypothesis

Higher seat classes, such as Business and Premium Economy Class, will result in higher booking amounts compared to Economy Class	Accepted
Booking trends may attract more business class travel on weekdays and more economy travel on weekends.	Rejected
Flights with a longer length of stay have a higher booking amount.	Rejected
Customers booking Business class tickets have a longer purchase lead time compared to Economy class tickets	Accepeted
Internet users tend to have higher booking amounts because they usually book round trips, while mobile users more often book one-way tickets.	Rejected
Customers who opt for more services will likely have higher booking amounts, as each additional service adds to the total cost	Accepted

Cutomer Preferrence Analysis

Around 74.5% of passengers opt for extra baggage, making it the most preferred add-on service, likely driven by long-stay and international travelers.





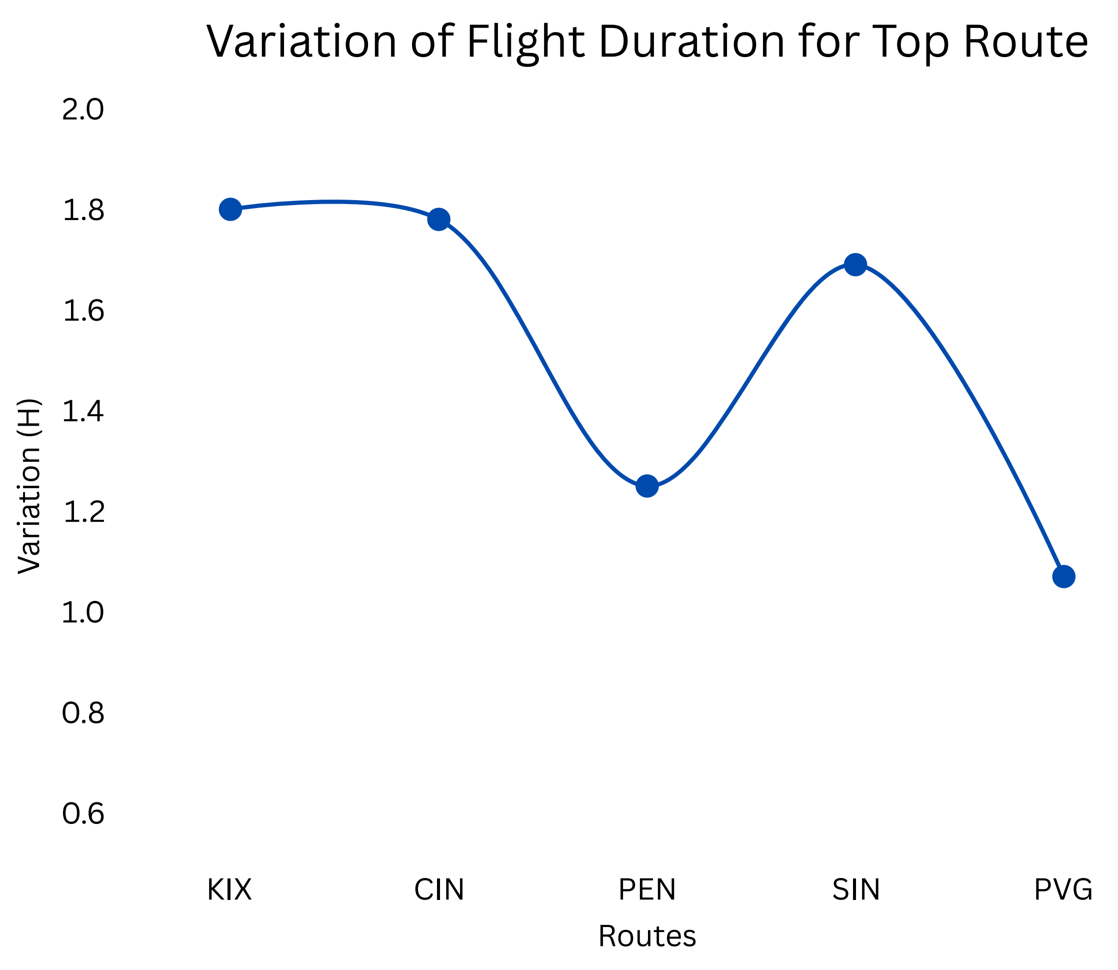
Bookings are almost evenly distributed across Economy (33.5%), Premium Economy (33.4%), and Business (33.1%), indicating balanced demand across customer segments.

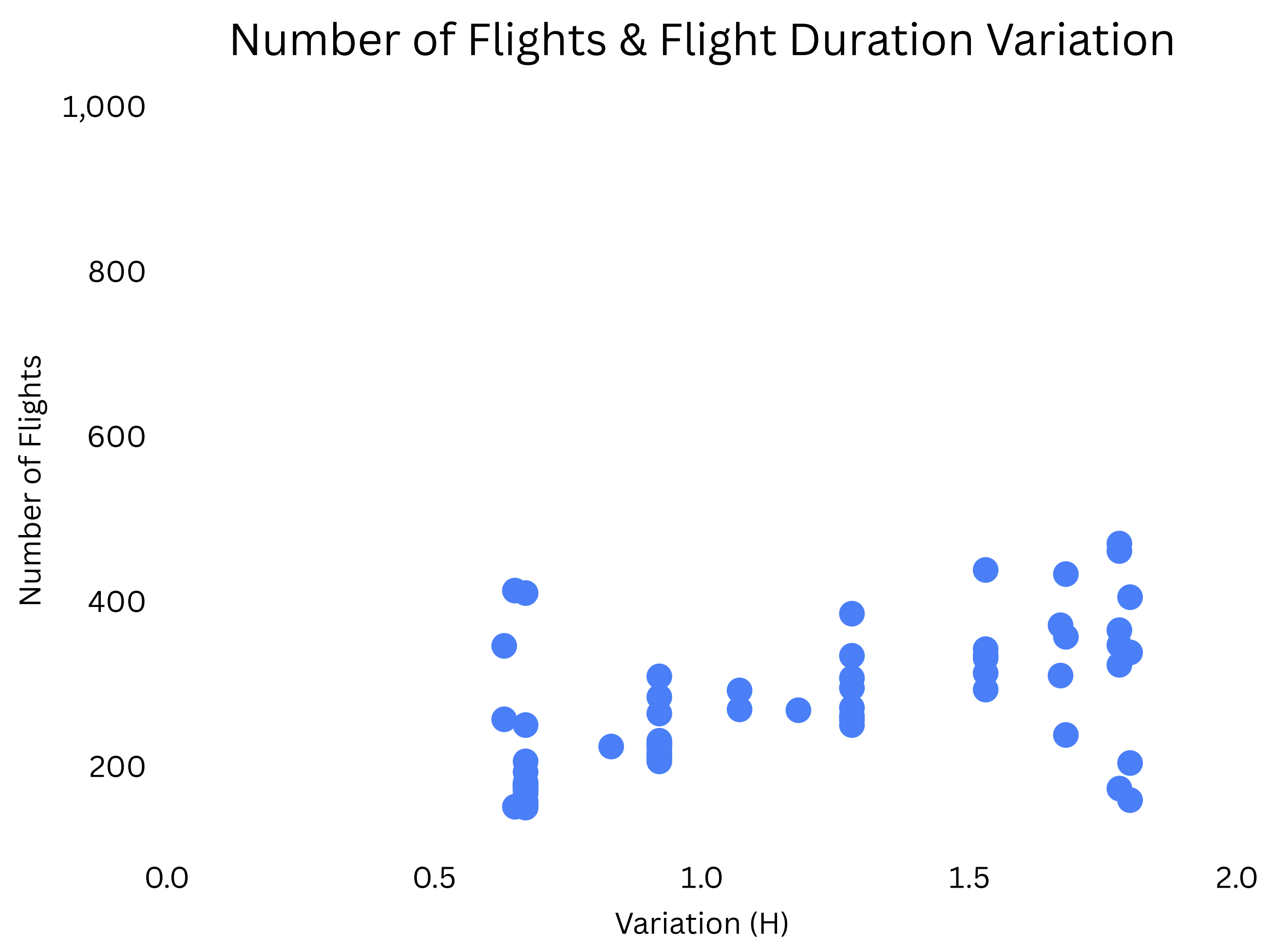
High adoption of ancillary services highlights strong upselling opportunities to increase non-ticket revenue and overall booking value.

Operational Efficiency Analysis

Routes like KIX, CIN, and SIN show the highest flight duration variation, indicating possible operational inefficiencies or route-specific disruptions.

Correlation analysis shows that flight duration variation is independent of the number of flights, suggesting delays are driven more by external operational factors than flight frequency.

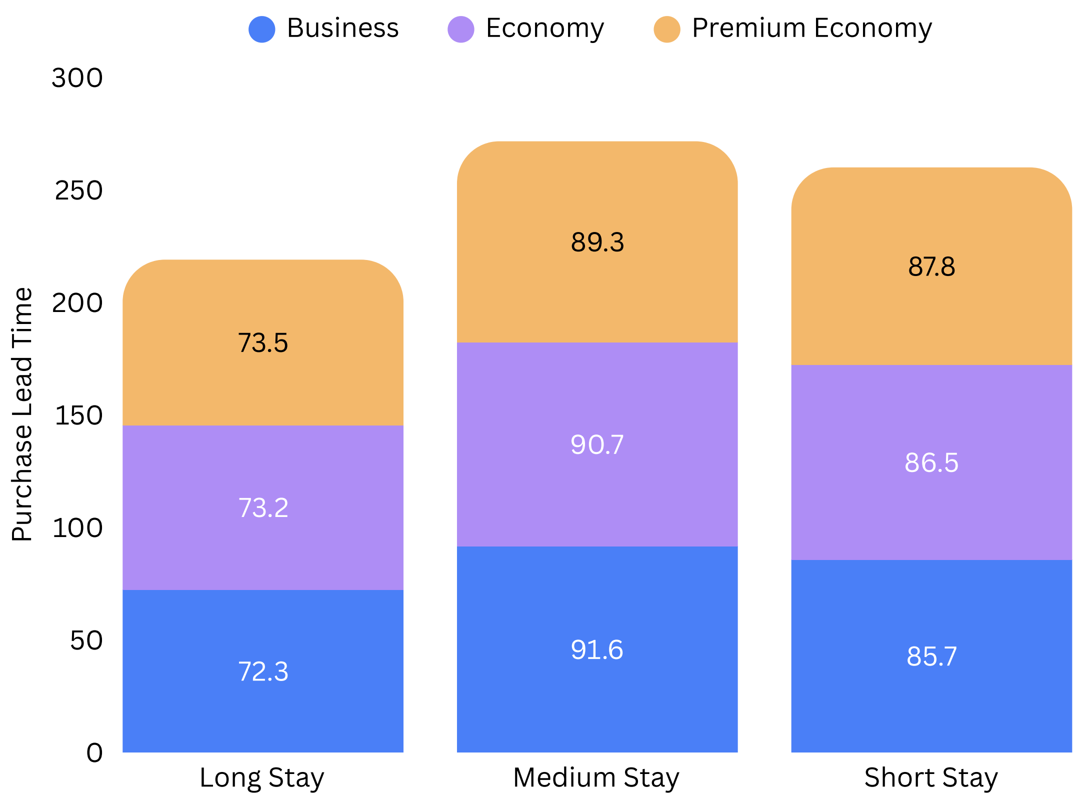




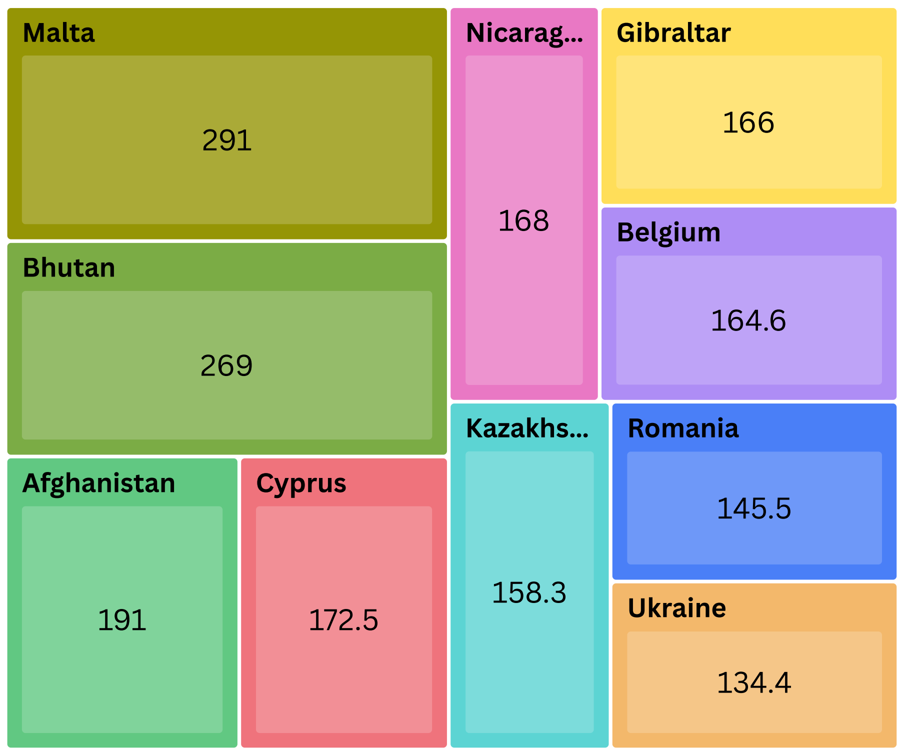
Indigo can improve operational efficiency by identifying route-specific causes such as air traffic congestion, weather conditions, or turnaround delays on high-variation routes.

Booking Trend Analysis

Booking origin such as Malta and Bhutan show significantly higher purchase lead times, suggesting advance travel planning for international or long-distance routes.



Route with Higher Purchase Lead Time

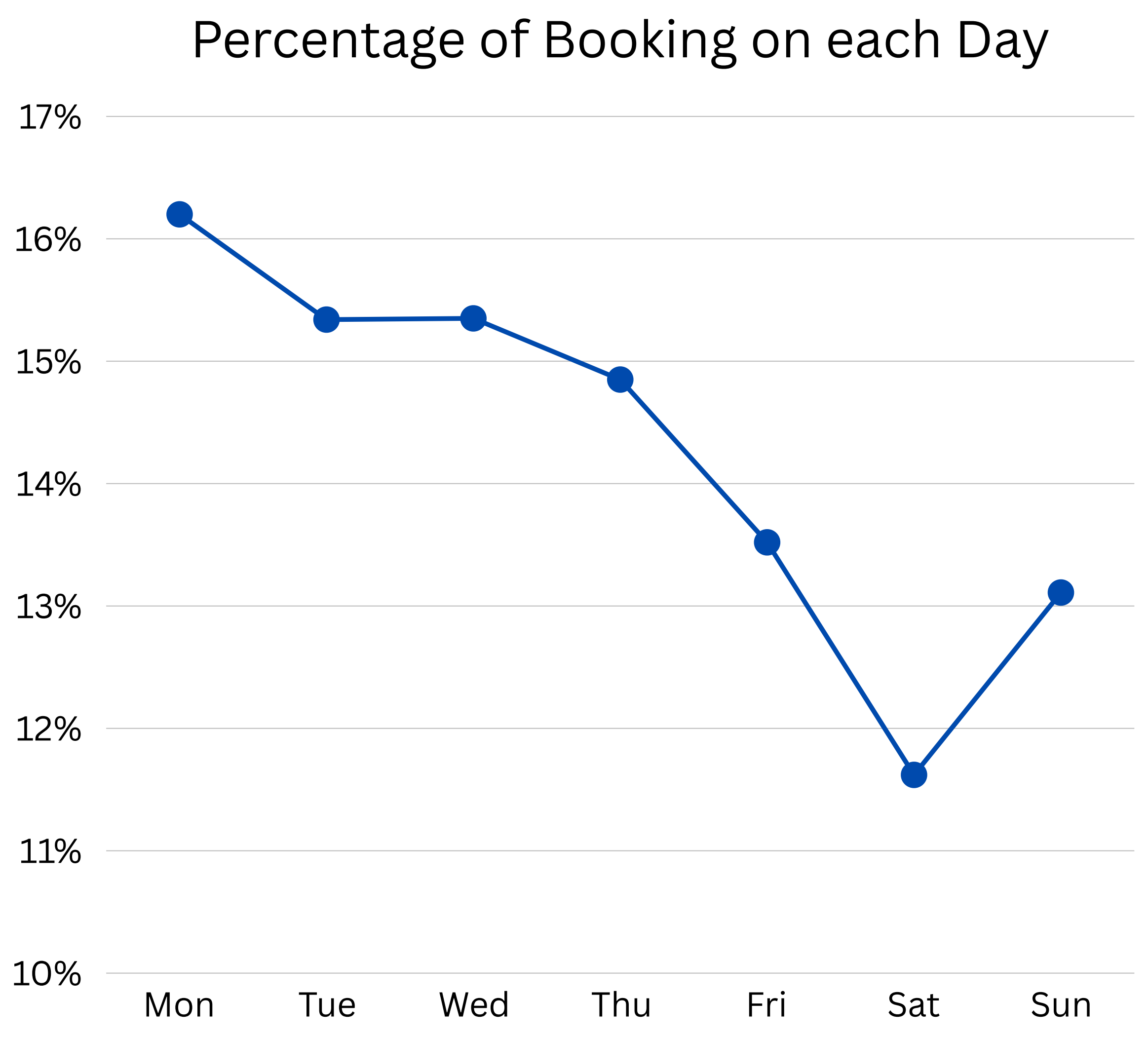


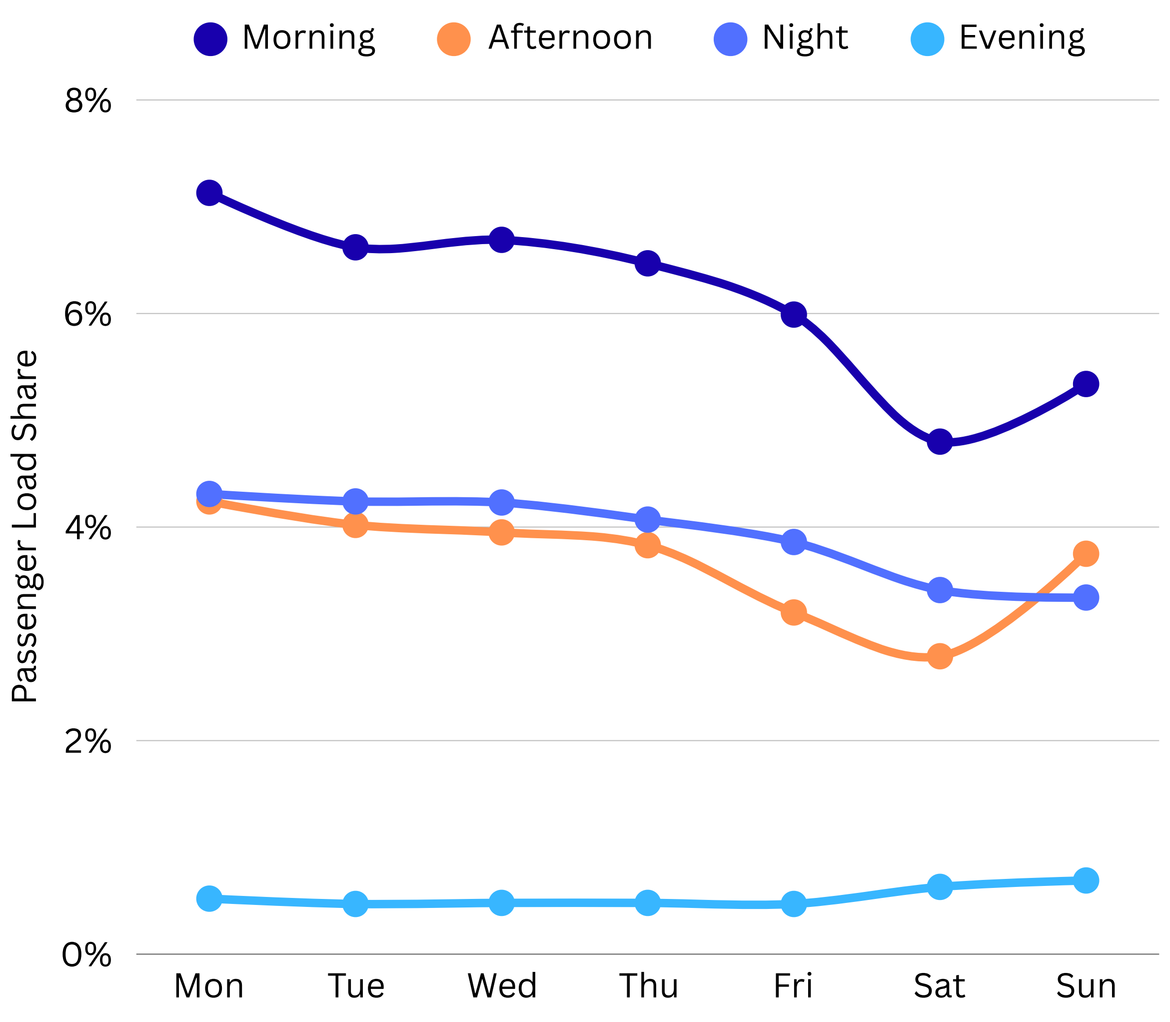
Average purchase lead time remains relatively consistent across stay types and seat classes, ranging between 70–90 days, indicating stable booking behavior.

High Purchase lead-time routes provide opportunities for early-bird pricing strategies, demand forecasting, and promotional campaigns to improve occupancy and revenue planning.

Passenger Load Analysis

Passenger load are higher during weekdays, especially on Monday and Tuesday, while morning flights consistently show the highest passenger load across all days. Weekend demand, particularly on Saturday, remains comparatively lower.

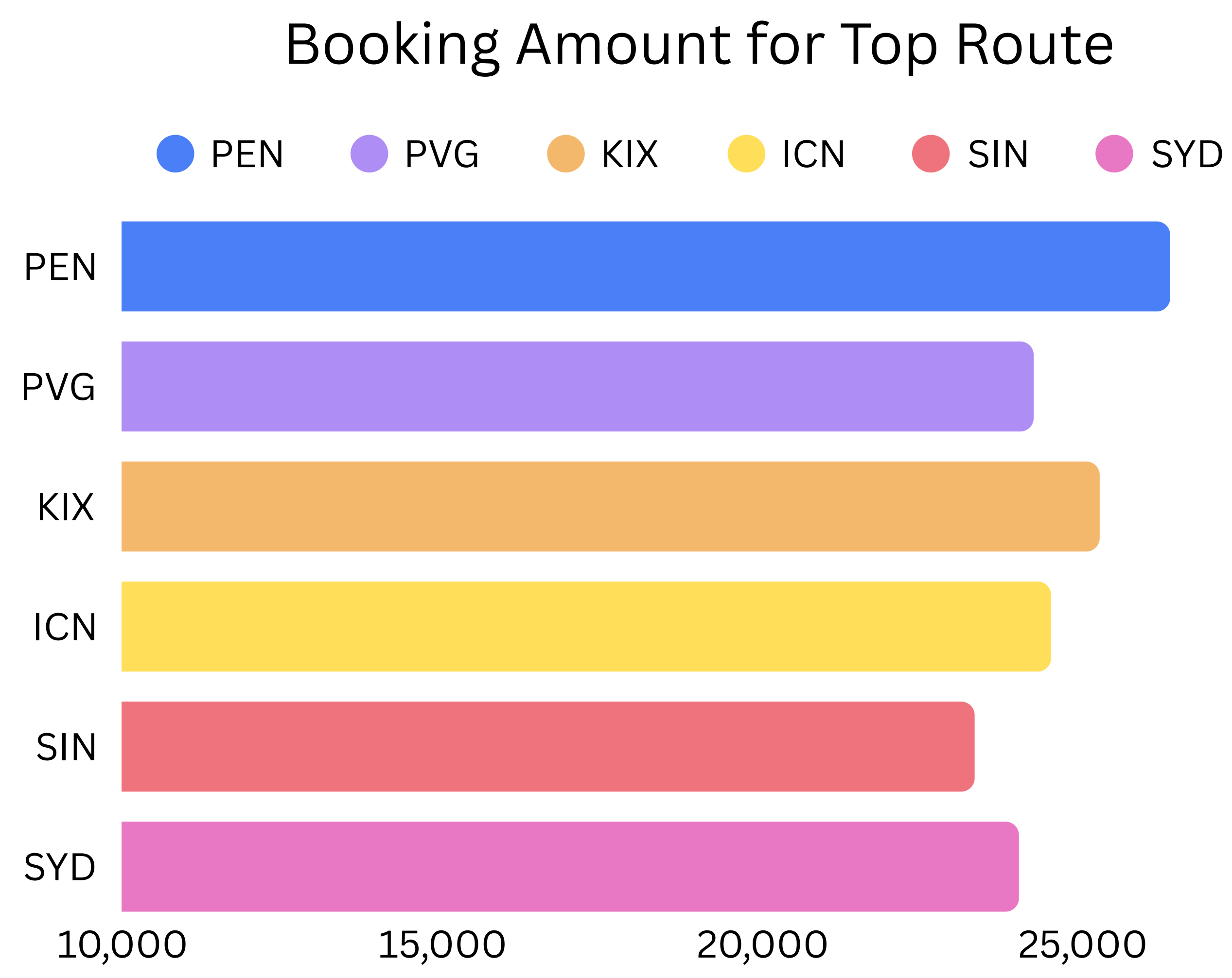


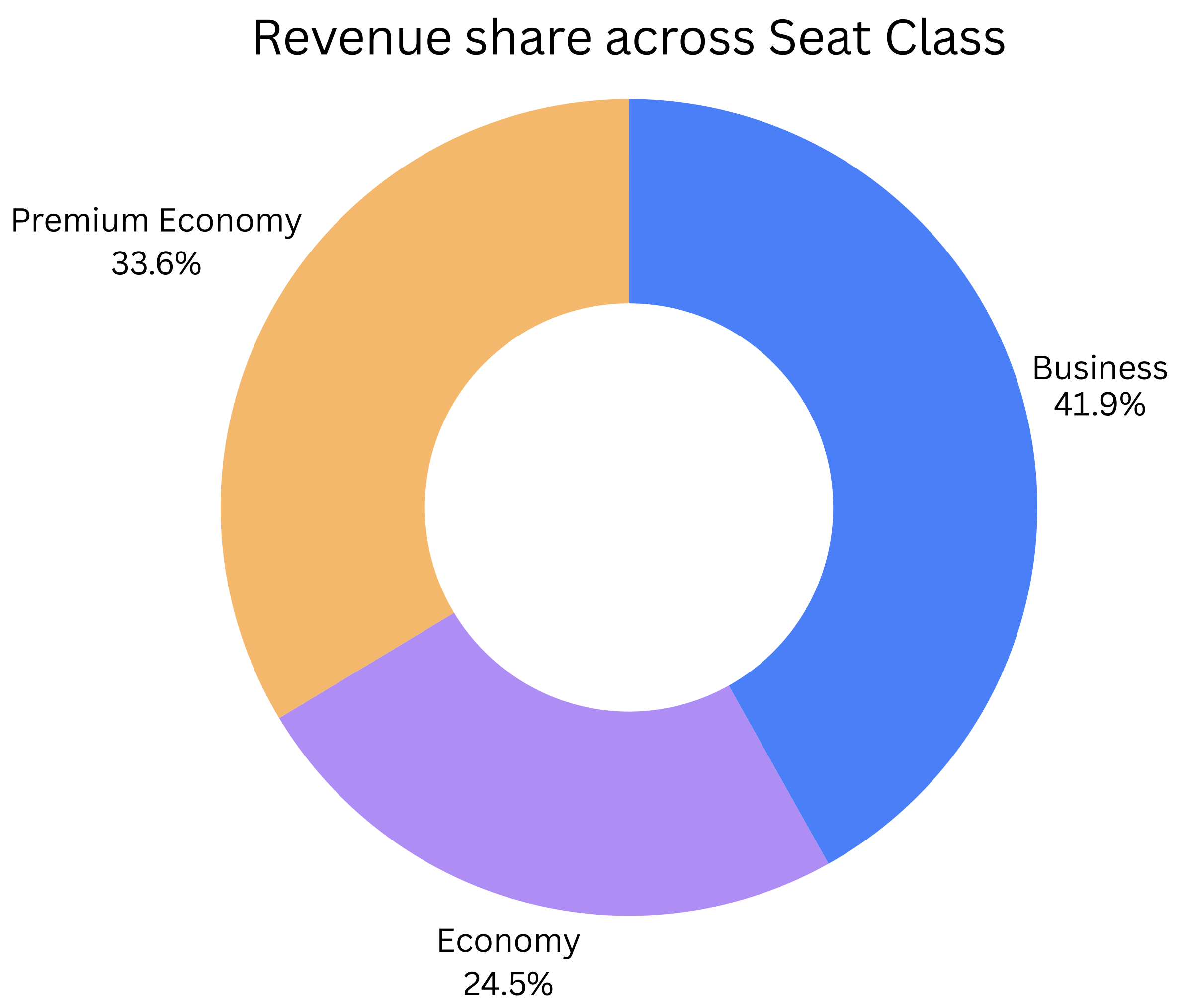


Higher weekday and morning demand is likely due to business and work-related travel preferences, where passengers prefer early departures for same-day meetings, office schedules, and operational convenience.

Revenue Distribution Analysis

Business class contributes the highest revenue share at 41.9%. Routes like PEN, KIX, and ICN generate the highest booking revenue, indicating strong premium and international route demand.





Higher revenue contribution from business class is likely driven by corporate and long-distance travelers willing to pay premium fares for comfort and flexibility. High-performing routes may reflect strong business connectivity, tourism demand, and higher passenger traffic.

Key Insights

Indigo Airlines can improve business outcomes by increasing focus on high-performing routes, strengthening Internet booking channels, and implementing targeted upselling strategies for ancillary services. Optimizing weekday flight schedules and personalized promotions can further improve booking conversion, occupancy rates, and overall route profitability while enhancing customer experience.

Route Profitability

Routes like PEN, PVG, and KIX shows high booking revenue with higher passenger volume and flight frequency, making them key profitability routes.

Customer Preference

Passengers opting for add-on services generate ~22K–28K booking value compared to ~16K for regular bookings, highlighting strong upselling potential.

Booking Trends

Nearly 90% of successful bookings come through internet sales channels, while overall booking conversion remains around ~15%.

Seasonal Analysis

Around 65–70% of passengers prefer weekday travel, and medium/short-stay travelers tend to book flights earlier.

THANK YOU